

Pre-Board Exam(2024-25)

Class-X

Time-2 Hours

Subject :Artificial Intelligence(417)

MM-50

General Instructions:

1. Please read the instructions carefully.
2. This Question Paper consists of 21 questions in two sections: Section A & Section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. Out of the given (5 + 16 =) 21 questions, a candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours.
5. All questions of a particular section must be attempted in the correct order.
6. **SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):**
 - i. This section has 05 questions.
 - ii. Marks allotted are mentioned against each question/part.
 - iii. There is no negative marking.
 - iv. Do as per the instructions given.
7. **SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):**
 - i. This section has 16 questions.
 - ii. A candidate has to do 10 questions.
 - iii. Do as per the instructions given.
 - iv. Marks allotted are mentioned against each question/part.

SECTION A: OBJECTIVE TYPE QUESTIONS

Q. 1	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)	
i.	Which of these are examples of positive feedback? (a) Excellent, your work has improve (b) I noticed, your dedication towards the project (c) You are always doing it the wrong way (d) All of the above	1
ii.	What is the term used when you quickly click the left mouse button twice? (a) Hover (b) Drag and drop (c) Double clicking (d) Moving	1
iii.	Srishti is a young woman who makes earrings. She buys jute from a farmer and makes earrings from that. She sees that most women in her village do not work. So, she hires two women to help her. As her orders increase, she hires three more women to work for her. How was she helping her village grow? (a) By selling earrings to women without a job (b) By purchasing earrings from the local market (c) By buying jute from the local farmer and by providing jobs to local women (d) By attracting the women in her village with her creative earrings	1
iv.	_____ is a series of postures and breathing exercises practiced to achieve control of body and mind. (a) Meditaion (b) Nature walk (c) Yoga (d) Drill	1
v.	The _____ conveys the message to the sender in the form of feedback for the completion of communication cycle. (a) Sender (b) Receiver (c) Channel (d) Protocol	1
vi.	Which of the following are common Sustainable Practices? a. Reusing of Waste Water b. Conserving Rain Water c. Segregating waste at source d. All of the above	1

	some basic queries that they are coded for and connect them to human executives once they are unable to handle the conversation.	
v.	A corpus contains 12 documents. How many document vectors will be there for that corpus? a. 12 b. 1 c. 24 d. 1/12	1

Q. 4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i	_____ is the sub-field of AI that is focused on enabling computers to understand and process human languages. a) Deep Learning b) Machine Learning c) NLP d) Data Sciences	
ii.	_____ helps in assigning a predefined category to a document, organize it in such a way that helps customers to find information they want. For example spam filtering in email, auto tagging on social media, categorization of news articles etc.	1
iii.	The makes the data understandable for humans as we can discover trends and patterns out of it. a) Random Data b) Graphical Representation c) Unstructured Data d) None of the above	1
iv	What does a pixel represent in an image? A) A segment of a video B) The smallest unit of an image C) The color depth of an image D) The brightness of an image	
v	Google Assistant, Alexa, Cortana, Siri are examples of a) Script Bot b) Smart Bot c) Sling Bot d) None of these	
vi.	What is the full form of TF-IDF?	1

Q. 5	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Which of the following statement is incorrect? a. Email Filtering is an application of NLP b. Price comparison websites are applications of Data Science c. Virtual Assistants are not applications of NLP d. Self-driving cars are applications of Computer Vision.	1
ii.	Which two evaluation methods are used to calculate F1 Score? (a) Precision and Accuracy (b) Precision and Recall (c) Accuracy and Recall (d) Precision, F1 score	1
iii.	What does the pixel value represent in a grayscale image? A) The color B) The intensity C) The contrast D) The size	1
iv.	Machine/computer understands the language of numbers (binary numbers 0's and 1's). Everything that is sent to the machine has to be converted to numbers. a) True b) False c) not d) None of these	1
v.	"Converting text to a common case" is a step in Text Normalization. (True/False)	1
vi.	Raunak was learning the conditions that make up the confusion matrix. He came across a scenario in which the machine that was supposed to predict an animal was always predicting not an animal. What is this condition called? (a) False Positive (b) True Positive (c) False Negative (d) True Negative	1

SECTION B: SUBJECTIVE TYPE QUESTIONS

Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks)

Answer each question in 20 – 30 words.

Q. 6	List of different type of verbal communication?	2
Q. 7	Sameera is always punctual at school. She has a regular schedule that she follows every day. She plans for study and play time in advance. Enlist the four steps Sameera must have followed for effective time management.	2
Q. 8	Enlist any two methods to protect our data on the computer.	2
Q. 9	What do entrepreneurs do when they run their business? Mention any two points.	2
Q. 10	Raj has a small convenience store in his locality. There are many other convenience stores in the area. Yet, Raj's store survives the competition and does well. Which stage of an entrepreneur's career process can you relate this to? Explain.	2

Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)

Q. 11	Write Examples of Features or applications used in Smart Phone that uses AI.	2
Q. 12	Explain Rule-based AI modelling approaches.	
Q. 13	Suhana works for a company wherein she was assigned the task of developing a project using AI project cycle. She knew that the first stage was scoping the problem. Help her list the remaining stages that she must go through to develop the project.	2
Q. 14	What is resolution, and how does it affect image quality?	2
Q. 15	What will be the results of conversion of the term, 'happily' in the process of stemming and lemmatization? Which process takes longer time for execution?	2
Q. 16	Create a confusion Matrix based on data given below True Positive : 150, True Negative : 60, False Negative : 50 and False Positive : 40	2

Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)

Q. 17	Samarth attended a seminar on Artificial Intelligence and has now been asked to write a report on his learnings from the seminar. Being a non-technical person, he understood that the AI enabled machine uses data of different formats in many of the daily based applications but failed to sync it with the right terminologies and express the details. Help Samarth define Artificial Intelligence, list the three domains of AI and the data that is used in these domains.	4
Q. 18	a) Define AI Project Cycle, Name all the stages of an AI Project cycle. b) Name the 4Ws of problem canvases under the problem scoping stage of the AI Project Cycle.	
Q. 19	Differentiate between rule-based and learning-based AI modelling approaches. Explain the following: (a) Supervised Learning (b) Unsupervised Learning	4
Q. 20	Samiksha, a student of class X was exploring the Natural Language Processing domain. She got stuck while performing the text normalisation. Help her to normalise the text on the segmented sentences given below: Document 1: Akash and Ajay are best friends. Document 2: Akash likes to play football but Ajay prefers to play online games.	4

Q. 21

Automated trade industry has developed an AI model which predicts the selling and purchasing of automobiles. During testing, the AI model came up with the following predictions.

CONFUSION MATRIX		REALITY	
		YES	NO
PREDICTED	YES	60	25
	NO	05	10

(i) How many total tests have been performed in the above scenario?

(ii) Calculate precision, recall and F1 Score.

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